Capstone Train, Analyze, Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach the full controlled capstone and protect final references.

Supply all development data and executable candidates;keep the teacher-only final evaluator behind the actual freeze gate.

Keep one habit: Use current comparison keys and separate matrix assessment;record actual review status rather than template approval.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Train and compare reproducible candidates on fixed development data

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Use the complete Week 34 synthetic parcel kit, reproduced in the student program: 40 training,20 validation and12 final feature rows with blank final labels. Compare baseline,V1 width-only and V2 width-plus-height on the same validation rows. Teacher final evaluator is private and used only after inspecting an actual saved freeze record. Other matrices are explicitly separate exercises,not this kit’s outcomes.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

The inspected set becomes development **evidence** for the revised version.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Opening retrieval

Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.

Look for: Define a feasible capstone question and acceptance criteria

Reference example, not the only acceptable wording: Teacher explanation: Compare a simple classifier with a **training**-majority baseline on the fictional parcel evaluation; no real handling or student decisions are supported.

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

Look for: Identify possible proxy features

Reference example, not the only acceptable wording: Teacher explanation: X occurs 8/10 north and 2/10 south in this sample. This supports an association, not a deterministic identity rule or a causal explanation.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Look for: Create features available at **prediction** time

Reference example, not the only acceptable wording: Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

End goal

By the end, I can explain capstone train, analyze, improve using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Capstone Dataset and Baseline

Proxy Features and Fairness

Feature Engineering

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

Concept to explain

One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.

Term	Meaning
baseline run	first reproducible model result
error audit	systematic review of mistakes
before/after	same evaluation around one change
regression	previously passing case now fails

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

▼ Teacher-only final key and evaluator;release after freeze

Teacher only:after verifying the actual saved V2 freeze record, run this evaluator or trace its 12 final references. Expected TN 3/FP 2/FN 0/TP 7 gives 10/12 accuracy, 7/9 precision and 7/7 recall; baselineTN5/FP 0/FN 7/TP 0 gives 5/12. This is the constructed parcel kit, not any separate worked matrix or a learner's Plant Classifier result.

Keep the key/evaluator in teacher materials until release; the example does not record an actual classroom sign-off. A revision chosen using these outcomes requires new final evidence.

```
import csv
from io import StringIO
from collections import Counter

# Complete student kit; final reference cells are deliberately blank.
DATA = """id,group_id,split,width_cm,height_cm,label
P00,P00,training,2,2,clear
P01,P01,training,3,2,clear
P02,P02,validation,4,2,clear
P03,P03,final,5,2,
P04,P04,training,6,2,clear
P05,P05,training,7,2,clear
P06,P06,validation,8,2,clear
P07,P07,final,9,2,
P08,P08,training,10,2,blocked
```

```
P09,P09,training,2,3,clear
P10,P10,validation,3,3,clear
P11,P11,training,4,3,clear
P12,P12,training,5,3,clear
P13,P13,validation,6,3,clear
P14,P14,final,7,3,
P15,P15,training,8,3,clear
P16,P16,training,9,3,blocked
P17,P17,validation,10,3,blocked
P18,P18,training,2,4,clear
P19,P19,training,3,4,clear
P20,P20,validation,4,4,clear
P21,P21,final,5,4,
P22,P22,training,6,4,clear
P23,P23,training,7,4,clear
P24,P24,validation,8,4,blocked
P25,P25,final,9,4,
P26,P26,training,10,4,blocked
P27,P27,training,2,5,clear
P28,P28,validation,3,5,clear
P29,P29,training,4,5,clear
P30,P30,training,5,5,clear
P31,P31,validation,6,5,clear
P32,P32,final,7,5,
P33,P33,training,8,5,blocked
P34,P34,training,9,5,blocked
P35,P35,validation,10,5,blocked
P36,P36,training,2,6,clear
P37,P37,training,3,6,clear
P38,P38,validation,4,6,clear
P39,P39,final,5,6,
P40,P40,training,6,6,blocked
P41,P41,training,7,6,blocked
P42,P42,validation,8,6,blocked
P43,P43,final,9,6,
P44,P44,training,10,6,blocked
P45,P45,training,2,7,clear
P46,P46,validation,3,7,clear
P47,P47,training,4,7,clear
P48,P48,training,5,7,blocked
P49,P49,validation,6,7,blocked
P50,P50,final,7,7,
P51,P51,training,8,7,blocked
P52,P52,training,9,7,blocked
P53,P53,validation,10,7,blocked
P54,P54,training,2,8,clear
P55,P55,training,3,8,clear
P56,P56,validation,4,8,blocked
```

```
P57,P57,final,5,8,
P58,P58,training,6,8,blocked
P59,P59,training,7,8,blocked
P60,P60,validation,8,8,blocked
P61,P61,final,9,8,
P62,P62,training,10,8,blocked
P63,P63,training,2,9,clear
P64,P64,validation,3,9,blocked
P65,P65,training,4,9,blocked
P66,P66,training,5,9,blocked
P67,P67,validation,6,9,blocked
P68,P68,final,7,9,
P69,P69,training,8,9,blocked
P70,P70,training,9,9,blocked
P71,P71,validation,10,9,blocked
"""
records = list(csv.DictReader(StringIO(DATA)))
for row in records:
    row['width_cm'] = int(row['width_cm'])
    row['height_cm'] = int(row['height_cm'])
train = [r for r in records if r['split'] == 'training']
valid = [r for r in records if r['split'] == 'validation']
final_features = [r for r in records if r['split'] == 'final']
assert len({r['id'] for r in records}) == len(records) == 72
sets = [{r['group_id'] for r in part} for part in [train, valid, final_features]]
assert not (sets[0] & sets[1] or sets[0] & sets[2] or sets[1] & sets[2])
assert all(not r['label'] for r in final_features)
counts = Counter(r['label'] for r in train)
constant = 'blocked' if counts['blocked'] > counts['clear'] else 'clear'
predictions = [constant] * len(valid)
correct = sum(p == r['label'] for p, r in zip(predictions, valid))
positives = sum(r['label'] == 'blocked' for r in valid)
found = sum(p == r['label'] == 'blocked' for p, r in zip(predictions, valid))
print('roles', len(train), len(valid), len(final_features))
print('training clear/blocked', counts['clear'], counts['blocked'])
print('frozen baseline', constant)
print('validation correct/total', correct, len(valid))
print('validation blocked found/total', found, positives)
print('final references absent: not scored')
# IDs/groups are audit metadata, not model features. Tie rule chooses clear.

from fractions import Fraction

# Declare the two candidates and selection rule before inspecting validation.
# V2 changes only the available feature set. k=1 and all tie rules stay fixed.
def fit_neighbour(training, features):
    if not training or not features:
        raise ValueError('Need training records and named features')
```

```
    if any(r['label'] not in ('clear', 'blocked') for r in training):
        raise ValueError('Training labels must be known')
    lows = tuple(min(r[f] for r in training) for f in features)
    spans = tuple(max(r[f] for r in training) - lo for f, lo in zip(features, lows))
    if any(span == 0 for span in spans):
        raise ValueError('This experiment requires nonconstant training features')
    def scale(row):
        return tuple(Fraction(row[f] - lo, span) for f, lo, span in zip(features, lows,
spans))
    stored = [(r['id'], scale(r), r['label']) for r in training]
    return features, lows, spans, stored

def predict(row, model):
    features, lows, spans, stored = model
    point = tuple(Fraction(row[f] - lo, span) for f, lo, span in zip(features, lows,
spans))
    # Manhattan distance; exact ties use the smallest training ID.
    nearest = min(stored, key=lambda item: (sum(abs(a-b) for a,b in zip(point,item[1])),
item[0]))
    return nearest[2]

def matrix(part, preds):
    if len(part) != len(preds) or not part:
        raise ValueError('Need one prediction per labelled case')
    cells = [0, 0, 0, 0]  # TN, FP, FN, TP; blocked positive
    for r, p in zip(part, preds):
        if r['label'] not in ('clear', 'blocked') or p not in ('clear', 'blocked'):
            raise ValueError('Unknown reference or prediction; do not score blank final
labels')
        cells[2 * (r['label'] == 'blocked') + (p == 'blocked')] += 1
    return cells

models = {'V1': fit_neighbour(train, ('width_cm',)),
          'V2': fit_neighbour(train, ('width_cm', 'height_cm'))}
validation_predictions = {v: [predict(r,m) for r in valid] for v,m in models.items()}
validation_matrices = {v: matrix(valid, ps) for v,ps in validation_predictions.items()}
# Predeclared: most validation correct; then fewer features; then version name.
chosen = min(models, key=lambda v: (-(validation_matrices[v][0]+validation_matrices[v]
[3]), len(models[v][0]), v))
print('validation baseline TN FP FN TP', matrix(valid, [constant]*len(valid)))
for v in models:
    train_preds = [predict(r,models[v]) for r in train]
    print(v, 'training correct/40', sum(p==r['label'] for p,r in zip(train_preds,train)))
    print(v, 'validation TN FP FN TP', validation_matrices[v])
    print(v, 'validation errors', [r['id'] for r,p in zip(valid,validation_predictions[v])
if r['label']!=p])
gains = [r['id'] for r,a,b in
zip(valid,validation_predictions['V1'],validation_predictions['V2']) if a!=r['label'] and
```

```
b==r['label']]
regressions = [r['id'] for r,a,b in
zip(valid,validation_predictions['V1'],validation_predictions['V2']) if a==r['label'] and
b!=r['label']]
print('V2 gains', gains, 'regressions', regressions)
print('frozen choice', chosen)
# Save this code/data/feature contract and these predictions before key release.
final_predictions = [predict(r,models[chosen]) for r in final_features]
print('frozen final prediction ledger', list(zip([r['id'] for r in final_features],
final_predictions)))
print('Final score withheld: teacher reference release and freeze check required.')

# TEACHER ONLY: run after verifying the saved freeze record.
FINAL_KEY = {'P03': 'clear', 'P07': 'clear', 'P14': 'clear', 'P21': 'clear', 'P25':
'blocked', 'P32': 'blocked', 'P39': 'clear', 'P43': 'blocked', 'P50': 'blocked', 'P57':
'blocked', 'P61': 'blocked', 'P68': 'blocked'}
assert set(FINAL_KEY) == {r['id'] for r in final_features}
final_labelled = [dict(r, label=FINAL_KEY[r['id']]) for r in final_features]
print('TEACHER frozen final TN FP FN TP', matrix(final_labelled, final_predictions))
print('TEACHER same-case baseline TN FP FN TP', matrix(final_labelled,
[constant]*len(final_labelled)))
# These results do not document an actual classroom sign-off or permit retuning.
```

MAIN LESSON · SECTION 2

Model the actual kit comparison before any final release

> Keep kit validation,separate illustrative test matrices and pictured leaf counts distinct.

Complete kit contract,validation and errors

The complete program repeats all 72 kit records. Each ID/group is one distinct fictional parcel. Training/validation/final roles are 40/20/12 and stay disjoint. V1 uses width_cm; V2 uses width_cm and height_cm. Scale each feature with its training minimum/range:width 2/8, height 2/7. Use Manhattan distance, k=1, and smallest training ID on an exact distance tie.

Fractions preserve exact arithmetic for these integer dimensions. IDs break ties only; they are not distance features. Choose most validation-correct, then fewer features, then version name. Declare this before reading results.

Same20 validation parcels; blocked positive · TN · FP · FN · TP · Correct /20 · Blocked found /11

> **Same20 validation parcels; blocked positive:** Training-majority clear baseline
>
> **TN:** 9
>
> **FP:** 0
>
> **FN:** 11
>
> **TP:** 0
>
> **Correct /20:** 9 =45%
>
> **Blocked found /11:** 0

> **Same20 validation parcels; blocked positive:** V1 width only
>
> **TN:** 9
>
> **FP:** 0
>
> **FN:** 7
>
> **TP:** 4
>
> **Correct /20:** 13 =65%
>
> **Blocked found /11:** 4

> **Same20 validation parcels; blocked positive:** V2 width and height
>
> **TN:** 9
>
> **FP:** 0
>
> **FN:** 3
>
> **TP:** 8
>
> **Correct /20:** 17 =85%
>
> **Blocked found /11:** 8

Case-level comparison	Complete saved record
V1 validation mistakes	P24, P42, P49, P56, P60, P64, P67
V2 fixes	P42, P49, P60, P67; four more correct, +20 percentage points

Case-level comparison	Complete saved record
V2 remaining mistakes	P24, P56, P64; all are missed blocked cases
V2 new mistakes among V1-correct cases	None on these 20 inspected validation records
Training accuracy, not final evidence	Baseline22/40, V1 30/40, V2 40/40; V2 can retrieve its own unique training feature pair.
Conclusion and limit	Select V2 under the declared validation rule. These are constructed, inspected development cases; final performance is still withheld.

Model final-release discipline

The teacher releases final-test labels only after checking the freeze record. Score baseline and candidate on the same planned rows. Save every **prediction**, denominator and failure.

Do not edit a model while reading labels and then present only the edited version as the original test.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone evaluation and revision.

Separate pictured16-case trade-off

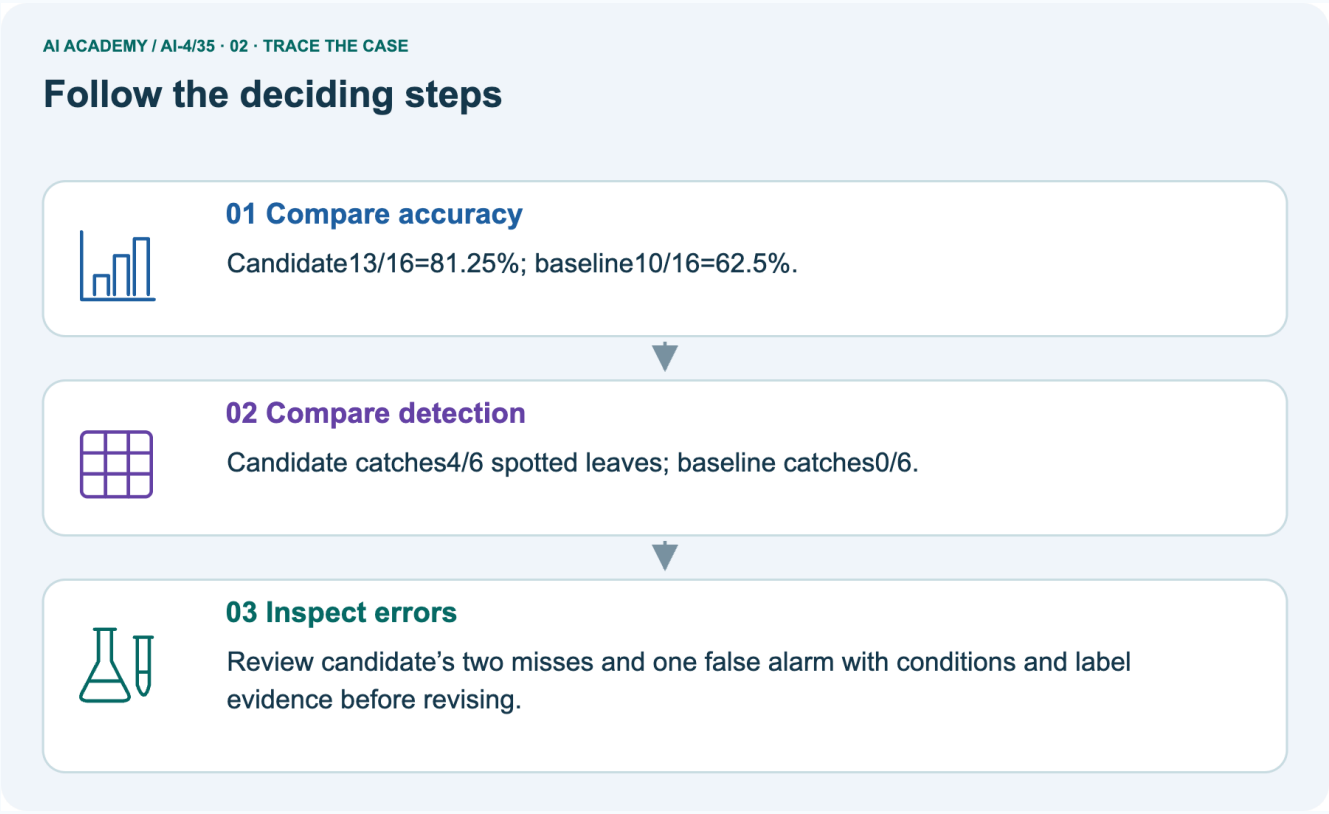
Choose the next capstone revision from explicit false alarms and misses.

Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate 16-case spotted-leaf arithmetic exercise, not the 72-row parcel kit:original TP 4, FN 2, FP 1, TN 9 gives 13/16 correct and 4/6 recall. The pictured revision fixes one miss but creates two false alarms:TP 5, FN 1, FP 3, TN 7 gives 12/16 correct and 5/6 recall. Higher recall and lower accuracy can coexist.

Which error matters depends on a declared task policy; these 16 cases become development evidence for a revision chosen from their errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare accuracy	Candidate13/16=81.25%; baseline10/16=62.5%.
2. Compare detection	Candidate catches4/6 spotted leaves; baseline catches0/6.
3. Inspect errors	Review candidate's two misses and one false alarm with conditions and label evidence before revising.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Change: A revision fixes one miss but creates two extra false alarms.

Prediction: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

▼ Separate illustrative12-case final matrix

Illustrative frozen test has 12 rows: candidate[[7, 1],[2, 2]], baseline[[8, 0],[4, 0]], blocked positive. Candidate accuracy 9/12=75 percent, precision 2/3 and recall 2/4; baseline accuracy 8/12 about 66.7 percent.

The candidate adds one false alarm and misses two rather than four blocked cases.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Term	Meaning
Final evaluation	Assessment of the frozen procedure on held-out evidence
Error audit	A documented inspection of mistakes and missing predictions
Revision log	A record of changes, reasons and affected evidence
Evidence status	Whether a result is development, untouched test or post-test reuse

Capstone evaluation and revision at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Frozen version	Saved before labels	Candidate v 1
First test	9 correct of 12	Report full error table
Proposed revision	Uses observed errors	Needs fresh evaluation

Compare the result with validation without claiming they should be identical. Check error types and coverage. A one-case accuracy difference on twelve rows supports a cautious classroom comparison, not a general reliability guarantee. Keep the original prediction ledger unchanged.

What does the one-correct-case improvement establish? Explain the steps or evidence behind your response.

▼ Earlier70/80validation illustration

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Plan improvement **evidence**.

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Train, audit, change one factor, and compare against the same validation set.

MODEL IDEA One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model v1 passes 14/20 validation cases; evening accuracy is 40%. The six failed cases are low-light cases.

One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Model v1 passes 14/20 validation cases; evening accuracy is 40%.

STEP 2 The six failed cases are low-light cases. One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

STEP 3 Add reviewed low-light **training** examples while freezing features and model settings.

STEP 4 Model v2 passes 17/20; evening rises to **70%**; inspect every old pass and record remaining limits.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Train, Analyze, Improve

The facts we will use

AI ACADEMY / AI-4/35

Candidate, before revision

	predicted spotted	predicted healthy
actual spotted	4	2
actual healthy	1	9

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/35

Capstone Train, Analyze, Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Compare accuracy

Candidate13/16=81.25%; baseline10/16=62.5%.

2. Compare detection

Candidate catches4/6 spotted leaves; baseline catches0/6.

3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

CHANGE ONE THING

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide a complete comparison,then remove help

> Collect a response before showing the key.

Current completion task and key

Using the complete parcel kit, explain the fixed learner and the single V1-to-V2 feature change. Reconstruct the validation matrices and all gained/regressed case IDs. Choose using the predeclared rule, then state what must be saved before final labels are released.

Fixed40 training/20 validation rows, k1, training-only min/range, Manhattan distance and ID tie rule. Only width→width+height changes. Baseline[9, 0, 11, 0]9/20, V1[9, 0, 7, 4]13/20, V2[9, 0, 3, 8]17/20 in TN/FP/FN/TP order. GainsP42/P49/P60/P67, no regressions; remaining missesP24/P56/P64.

Choose V2; save code/data versions, features/scales/ties, split IDs, predictions, selection rule and actual freeze/review status. Final score unknown to student until teacher release.

Current individual evidence-status check

V2 is selected using 17/20 validation and its three mistakes. Can the same 20 rows now be described as an untouched final test for V2? Explain the evidence status.

No. These20 validation records informed selection and error analysis. Freeze the chosen system and preserve the separate reserved final gate.

Separate pictured error-trade-off reasoning

Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

It catches more spotted leaves but raises false alarms and lowers accuracy. Report both trade-offs and decide using predeclared use costs. Changed-case check: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Changed-case reasoning: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Write the question to explain: Plan improvement **evidence**.

Reveal after **prediction**s: Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

Use the earlier transfer question as additional practice. Reserve the selected separate12-case frozen matrix for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: It establishes a difference on these 12 examples under the saved procedure, not reliable improvement for every future population.

Exit response: Version 1 frozen results, reason for change, reused-**evidence status** and the fact that version 2 has no new independent **final evaluation**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Use the prerequisite question: What changes after final-test labels inform a revision? Check the explanation before proceeding.
7 to 18 minutes	Verify each team freeze record before releasing final labels. Review the worked example and separate illustrative numbers from actual project evidence.
18 to 32 minutes	Teams run or trace frozen baseline and candidate predictions on all planned test cases. Save the unchanged prediction ledger and account for failures.
32 to 45 minutes	Teams calculate metrics and inspect error types. Use W1–W3 to check calculations and evidence status before writing claims.
Session 2 start to 8 minutes	Retrieve the distinction between final evaluation and error audit. Invite a counterexample to an overconfident claim.

Stage	Teacher action
8 to 20 minutes	Teams write an error audit and propose one follow-up tied to an observed limit. Keep version1 results unchanged.
20 to 35 minutes	Collect W4 independently. Teams then update their model card and clearly mark any revision as untested or supported only by reused development evidence.
35 to 45 minutes	Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: What does the one-correct-case improvement establish?

Teacher check: Teacher explanation: It establishes a difference on these 12 examples under the saved procedure, not reliable improvement for every future population.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record.

For this week: Candidate accuracy 9/12=75 percent, precision 2/3, recall 2/4; baseline 8/12 about 66.7 percent, with all four blocked cases missed.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Fixed40 training/20 validation rows, k1, training-only min/range, Manhattan distance and ID tie rule. Only width→width+height changes. Baseline[9, 0, 11, 0]9/20, V1[9, 0, 7, 4]13/20, V2[9, 0, 3, 8]17/20 in TN/FP/FN/TP order. GainsP42/P49/P60/P67, no regressions; remaining missesP24/P56/P64.

Choose V2; save code/data versions, features/scales/ties, split IDs, predictions, selection rule and actual freeze/review status. Final score unknown to student until teacher release. Selected matrix TN 6/FP 2/FN 1/TP 3:9/12 accuracy, 3/5 precision, 3/4 recall; baselineTN8/FN 4 gives 8/12 and misses all positives. Do not confuse that exercise with actual kit final labels.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Capstone Gate 2: reproduce v1, cluster errors, approve one change, rerun fixed cases, and publish all gains and **regression**s.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record.

For this week: Candidate accuracy 9/12=75 percent, precision 2/3, recall 2/4; baseline 8/12 about 66.7 percent, with all four blocked cases missed.

A stronger claim makes a stronger project: Version 1 frozen results, reason for change, reused-**evidence status** and the fact that version 2 has no new independent **final evaluation**.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Candidate accuracy 9/12=75 percent, precision 3/5=60 percent, recall 3/4=75 percent.

Baseline FP 0 FN 4 versus candidate FP 2 FN 1; baseline accuracy 8/12.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected separate12-case frozen matrix for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A higher overall score proves every important aspect improved.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess comparison reasoning and the fresh frozen matrix

> Current kit keys and independent matrix keys refer to separate fixtures.

Selected independent task and checked key

Selected case: Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

Teacher explanation: Candidate accuracy 9/12=75 percent, precision 3/5=60 percent, recall 3/4=75 percent. Baseline FP 0 FN 4 versus candidate FP 2 FN 1; baseline accuracy 8/12.

Current W1–W3/W5 keys

Current W1:only feature set changes; 40/20 membership, k1, training-only scaling, Manhattan and ID ties stay fixed. W2:baseline 9/20, V1 13/20, V2 17/20; four gains, no regressions, three misses; choose V2 by predeclared rule. W3:validation informed selection, so it is development evidence.

W5:save exact code/data/features/preprocessing/predictions and actual freeze status before key release; retain frozen final results and version later revisions. Selected W4 remains its own 12-case matrix, not a kit outcome.

Judge trace,scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

Additional transfer practice question

A revision raises overall validation accuracy from **80%** to 84% but lowers dim-light accuracy from **60%** to 40%. How should the improvement claim change?

Reasoned practice answer

Report both results with counts and evaluate against intended-use criteria. The overall gain does not justify hiding the subgroup **regression**; investigate or limit use.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any increase in overall accuracy counts as improvement for every user and condition.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. Reevaluate both versions on the same declared validation cases, including the hard cases unless an independently justified exclusion applies to both.

Record exact changes and per-case **prediction**s, then evaluate the selected version on untouched final data.

Explanations for the additional workbook examples

W1 Score the frozen test

Compute the worked candidate metrics and baseline accuracy.

Teacher explanation: Candidate accuracy 9/12=75 percent, precision 2/3, recall 2/4; baseline 8/12 about 66.7 percent, with all four blocked cases missed.

W2 Audit a difference

What does the one-correct-case improvement establish?

Teacher explanation: It establishes a difference on these 12 examples under the saved procedure, not reliable improvement for every future population.

W3 Label a revision

A threshold changes after test errors are read. Label its score on the same rows.

Teacher explanation: Post-test development or reused-test **evidence** for the revised version, not an untouched final test.

W4 Apply independently

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

Teacher explanation: Candidate accuracy 9/12=75 percent, precision 3/5=60 percent, recall 3/4=75 percent. Baseline FP 0 FN 4 versus candidate FP 2 FN 1; baseline accuracy 8/12.

W5 Update the card

What must remain visible when version 2 is proposed without a fresh test?

Teacher explanation: Version 1 frozen results, reason for change, reused-**evidence status** and the fact that version 2 has no new independent **final evaluation**.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **final evaluation** and **error audit** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Candidate accuracy 9/12=75 percent, precision 3/5=60 percent, recall 3/4=75 percent. Baseline FP 0 FN 4 versus candidate FP 2 FN 1; baseline accuracy 8/12.

Fresh retry candidate[[5, 1],[2, 2]] on 10 cases: accuracy 7/10, precision 2/3, recall 2/4. Require a test-status statement before any proposed tuning. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 36: ML Builder Expo. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Machine learning project defense. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Run a frozen **final evaluation** with complete accounting

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: What does the one-correct-case improvement establish?

Reasoned practice answer: Teacher explanation: It establishes a difference on these 12 examples under the saved procedure, not reliable improvement for every future population.

Check every learner before discussion: Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record.

For this week: Candidate accuracy 9/12=75 percent, precision 2/3, recall 2/4; baseline 8/12 about 66.7 percent, with all four blocked cases missed.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A revision fixes one miss but creates two extra false alarms. Require a **prediction** and a reason before comparing. Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

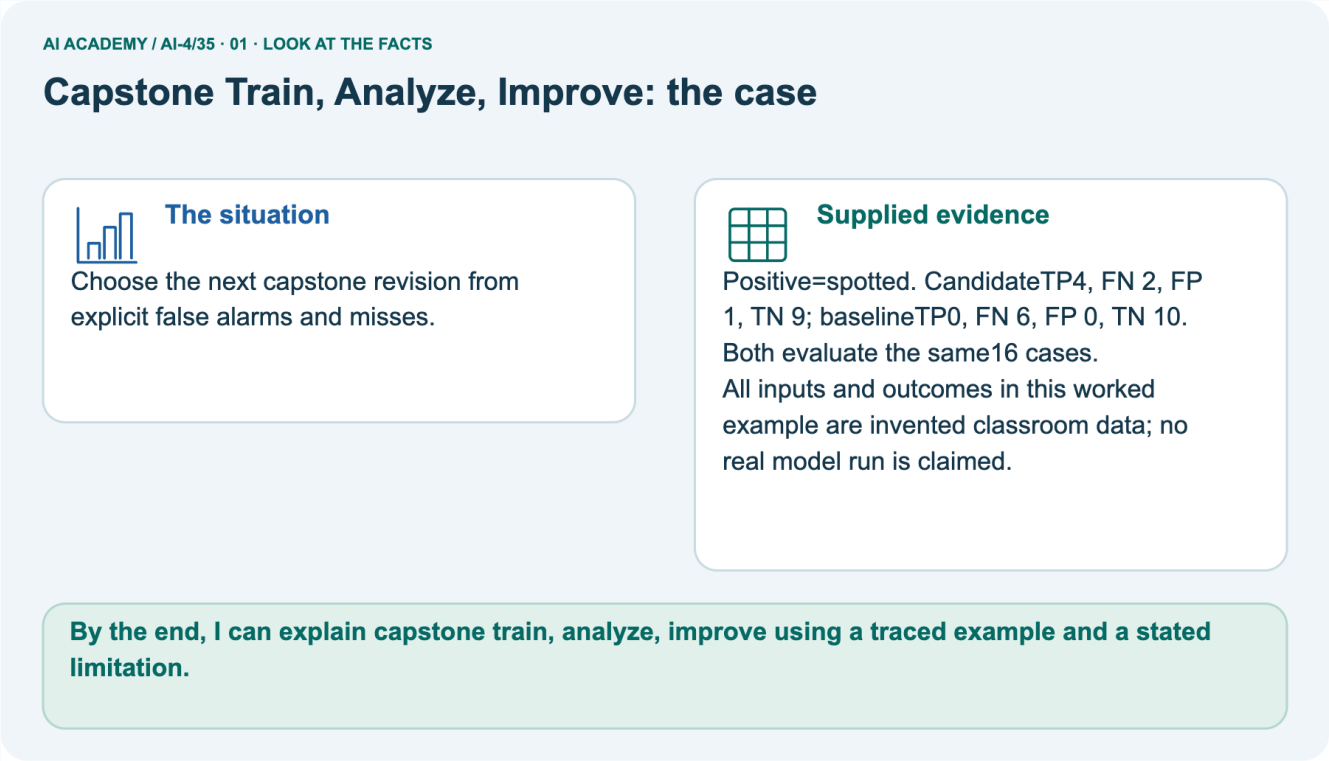
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Capstone Train, Analyze, Improve: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

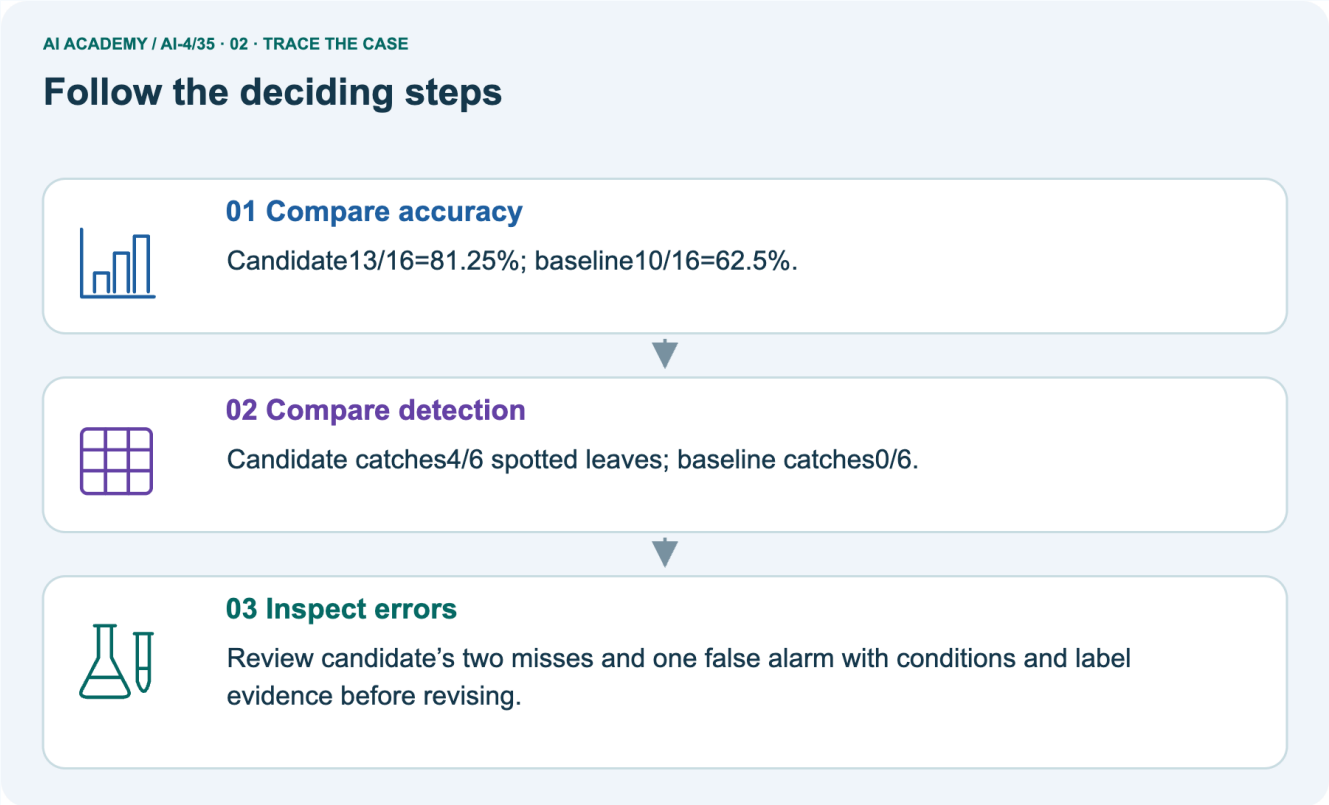
Choose the next capstone revision from explicit false alarms and misses. Supplied facts: Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



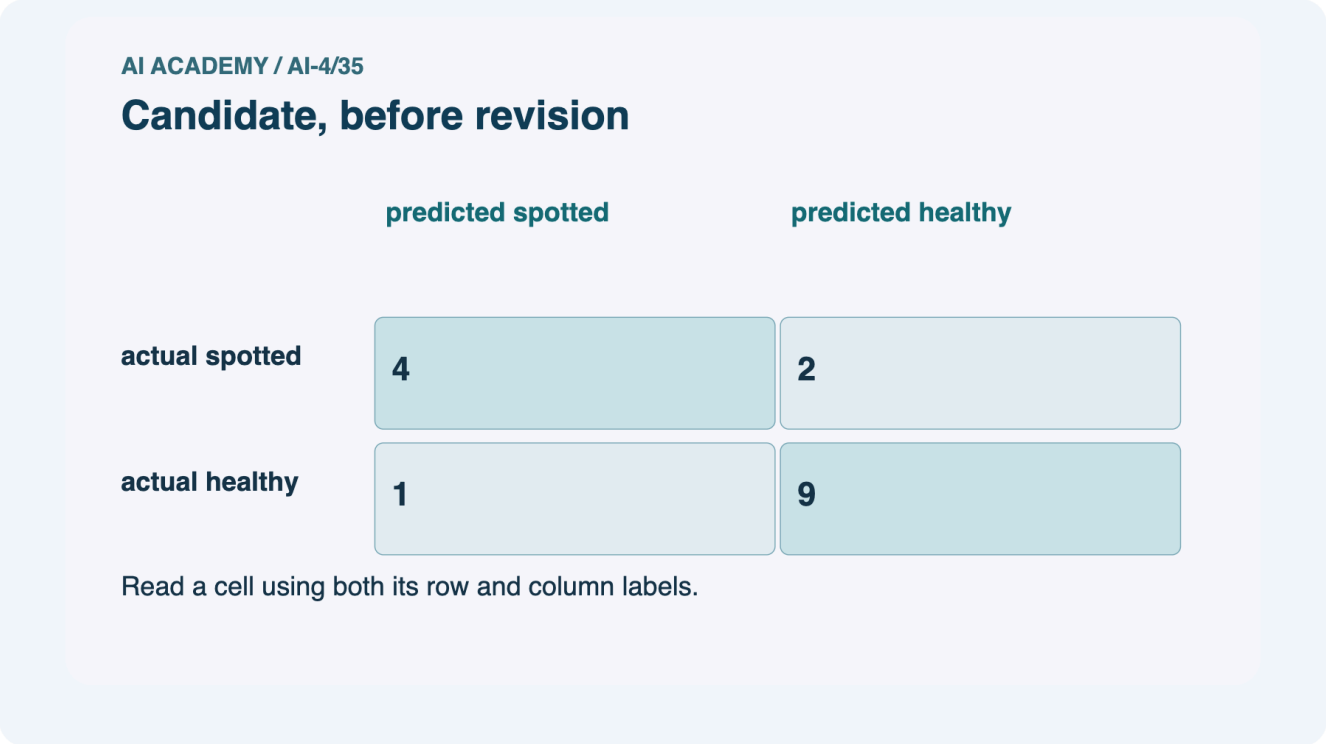
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare accuracy: Candidate13/16=81.25%; baseline10/16=62.5%. → Compare detection: Candidate catches4/6 spotted leaves; baseline catches0/6. → Inspect errors: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Candidate, before revision



Follow the labelled picture with the matching explanation. Open the vector diagram

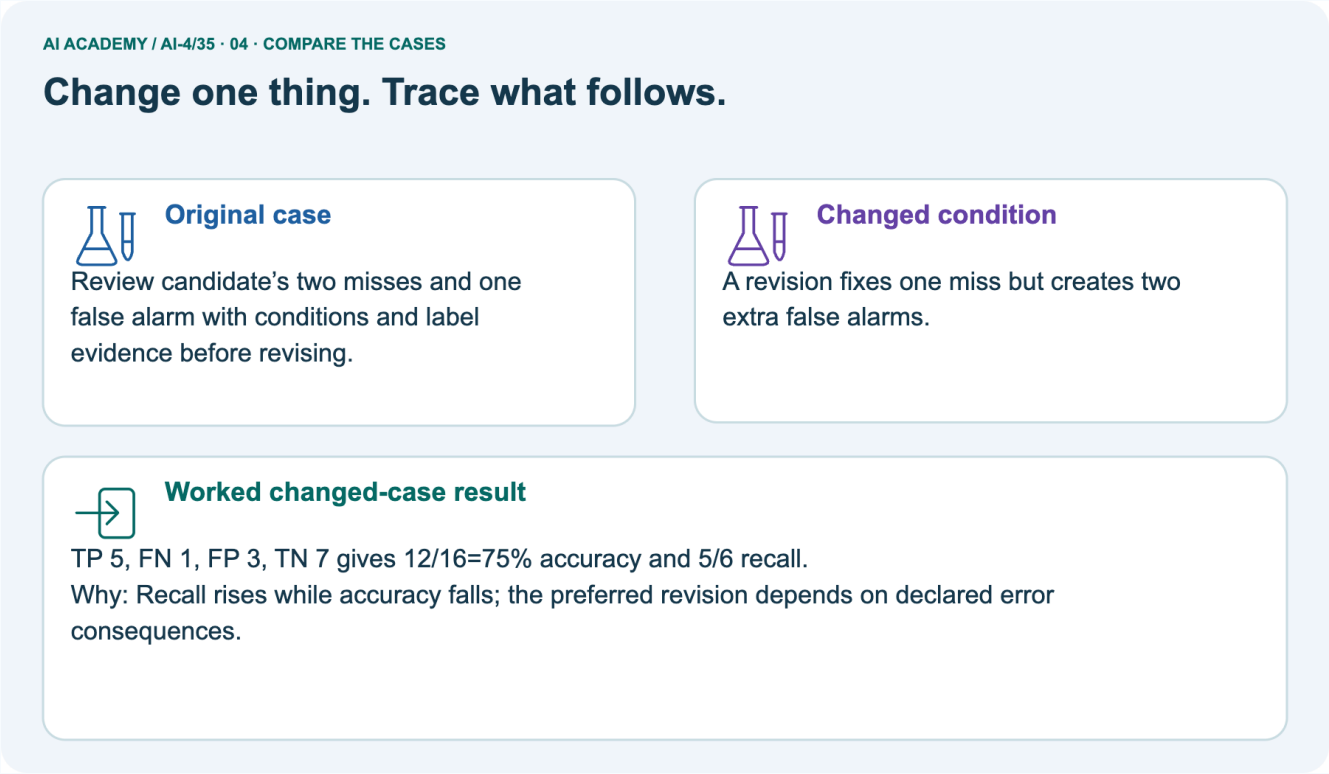
Candidate, before revision. actual spotted: predicted spotted = 4, predicted healthy = 2. actual healthy: predicted spotted = 1, predicted healthy = 9.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Changed condition: A revision fixes one miss but creates two extra false alarms.

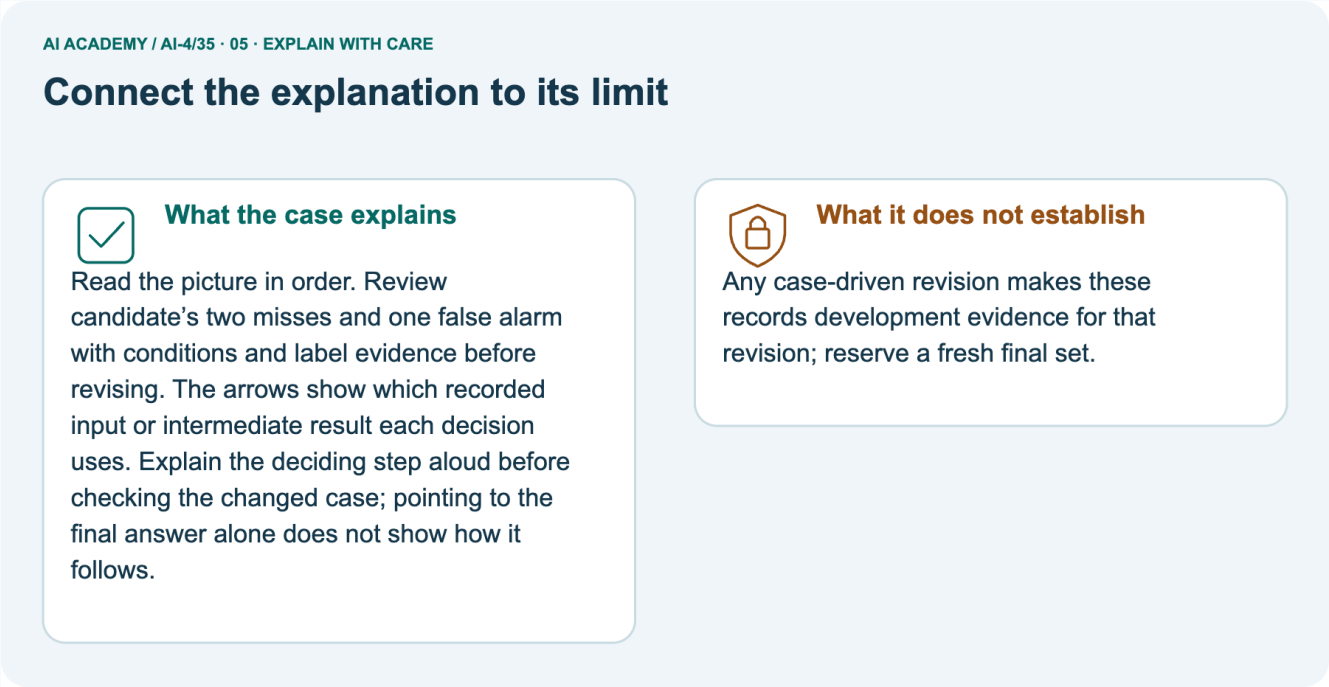
Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

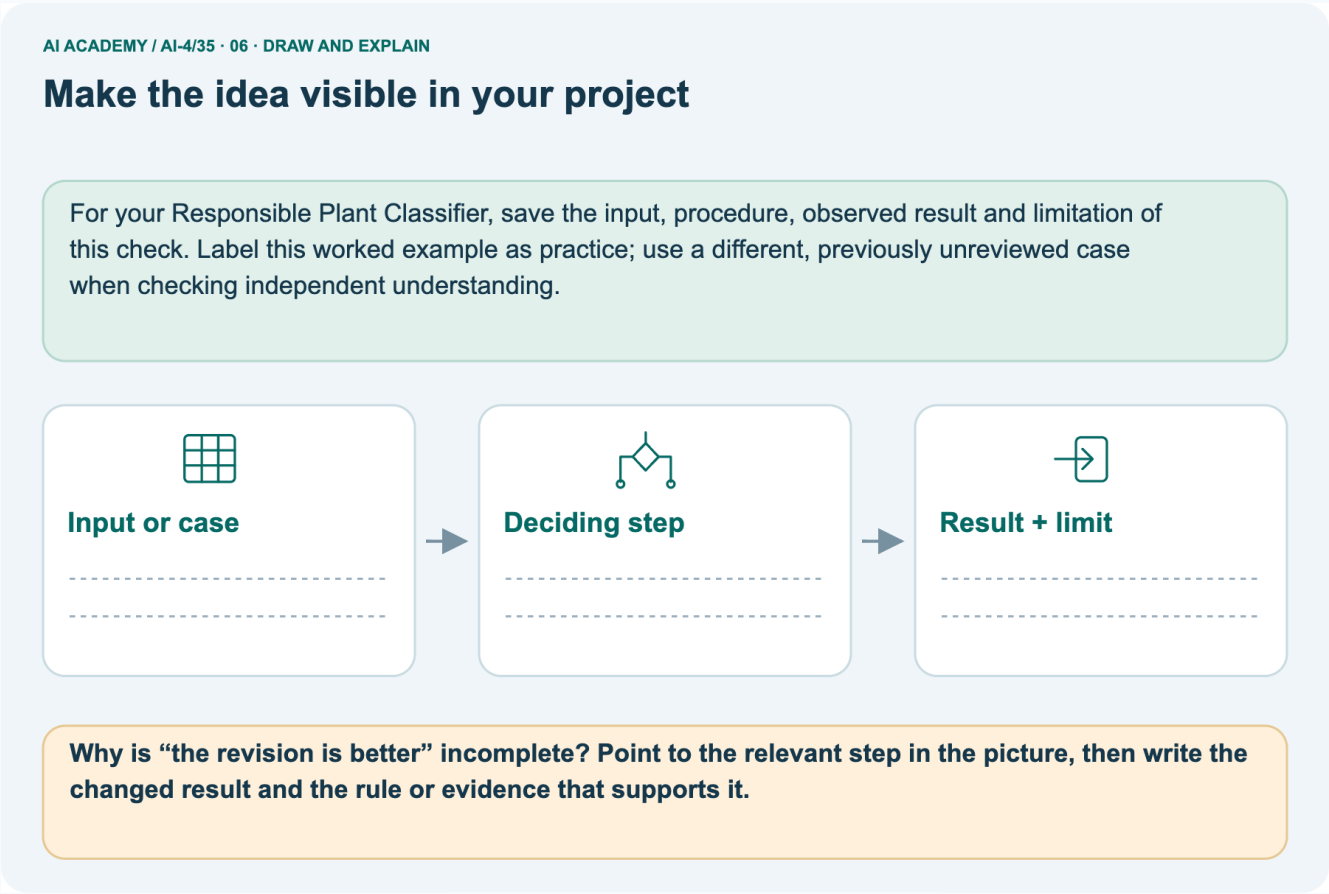
Limit: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It catches more spotted leaves but raises false alarms and lowers accuracy. Report both trade-offs and decide using predeclared use costs.

Changed-case check: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone Train, Analyze, Improve: the case

02. Follow the deciding steps

03. Candidate, before revision

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.